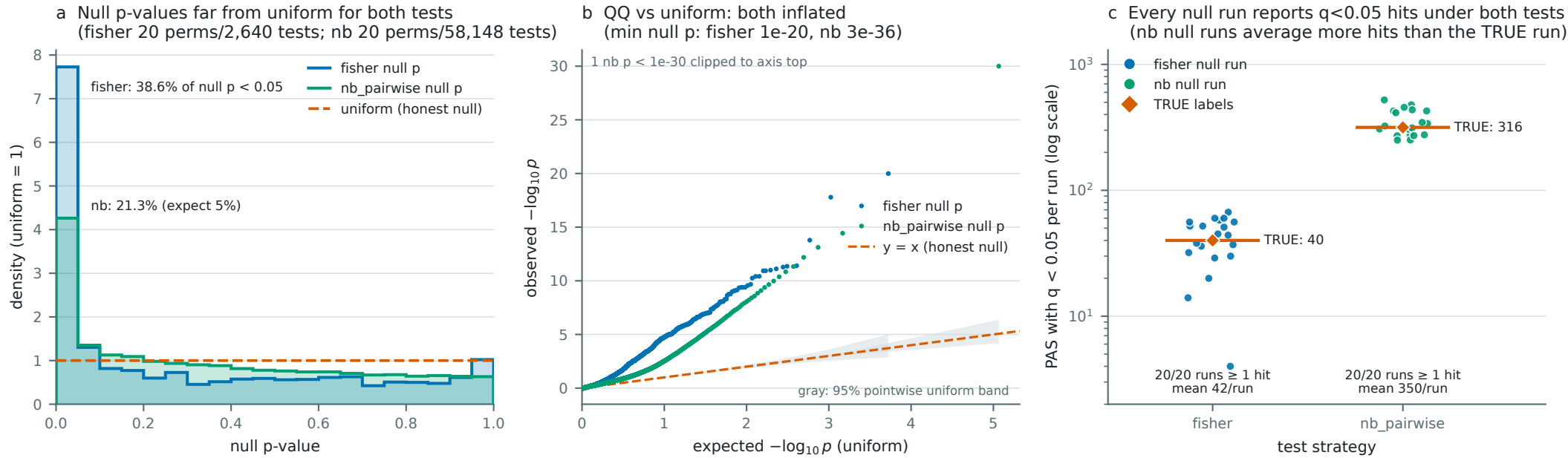


Neither `switch diff` strategy is FDR-calibrated under a stage-label permutation null -- fisher badly, nb_pairwise ~4x nominal -- Laughney TREG, 3,790 cells, 6 stage pairs per run



Null = obs['stage'] shuffled across cells (label counts kept), full pipeline re-run per permutation incl. top-200 marker pre-selection (double-dip deliberately inside the null), seeds 1-20; fisher 20/20 and nb 20/20 permutations available when drawn. BH families are per stage pair: fisher 112/120 and nb 118/120 null families contain ≥ 1 $q < 0.05$ hit (expect ~5%). Fisher drivers: the test treats reads as independent (pseudoreplication; reads within a cell are correlated) and Fisher p-values are discrete. nb_pairwise is a per-cell NB GLM (Wald z on the cluster coefficient, offset log library size) so it fixes pseudoreplication, yet 21% of null $p < 0.05$ and 12% of null $q < 0.05$ remain: per-PAS plug-in dispersion (clipped to $[1e-4, 10]$) hits the $1e-4$ floor in 21% of null tests -- near-Poisson variance -- and those tests carry ~50% of null $p < 0.05$; no dispersion shrinkage; marker double-dip included. Hit counts per run are not comparable across TRUE vs null: permuted labels homogenize clusters, so more PAS pass the ≥ 10 -expressing-cells-per-group filter (null nb runs test ~2,907 PAS/run vs 331 in TRUE); the per-test rates above are the calibrated quantities. Verdict: nb_pairwise q-values are also anti-conservative and cannot be taken at face value -- manuscript significance needs D4 cells-mode or permutation-calibrated thresholds.